ONLINE SHOPPERS PURCHASE INTENTION

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PROBLEM

Online shopping sites often suffer from the issue of **window shopping,** where a lot of people visit websites every day but only some people make a purchase. Understanding which kinds of people are likely to make a purchase will help companies to do the following-

- Improve targeted marketing
- Personalize user experiences
- Optimize website design

Problem Statement

The aim is to develop a machine learning model that can analyze online browsing behavior and predict whether a website visitor will complete a purchase during their session. This will be a great help to the companies.

OBJECTIVES

1. Analyze customer data to understand patterns that influence purchase decisions.
2. Build a machine learning model to predict whether a session will result in a purchase.
3. Train classification models to categorize sessions into Purchase and No Purchase.
4. Evaluate the performance of the model using metrics such as Accuracy and F1 score.
5. Identify factors that most strongly influence online purchasing behavior.

APPROACH

The dataset used has been taken from UCI Irvine's Machine Learning Repository. It contains 18 variables that affect whether a potential customer actually ends up making a purchase or not. Each record within the database covers only one browsing session. Some features of this dataset include-

- Region
- Browser
- OperatingSystems
- Weekend
- Month

Data preprocessing must be done, like changing T/F values to binary. After the data preprocessing, the data must be clustered into certain categories before machine learning models can be performed As this is a Binary Classification problem, Logistic Regression will be the best option available to us. Model performance will be evaluated with metrics like the **F1 Score,** a metricd which are important as it will help in evaluating the suitability of the model.

PROCESS FLOWCHART

1	2	3	4	5	6
Load UC Irvine Dataset	Data Preprocessing	Feature Analysis	Clustering and Model Training	Model Evaluation	Purchase Prediction

INSIGHTS AND IMPACTS

Personalized Marketing

Businesses can target users who are more likely to purchase with personalized advertisements and promotions.

Improved Website Design

Understanding user behavior can help optimize website layouts and product recommendations.

Supporting Small Businesses

Predictive models can help small businesses, which primarily operate outside of traditional brick and mortar models.

Better Customer Experience

Companies can offer customized shopping experiences based on user behavior patterns.